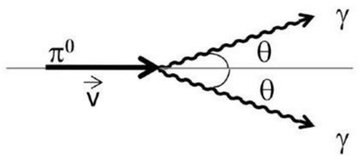
sPHENIX Electromagnetic Calorimeter Energy Scale Stability During RHIC Run 24

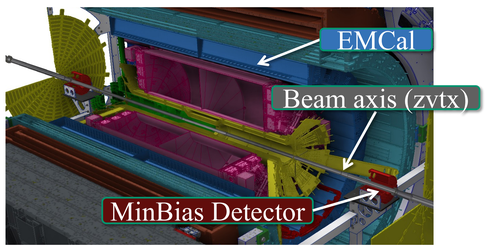
Elena R. Adams, Department of Physics, University of Colorado Boulder, for the sPHENIX Collaboration

Introduction

- sPHENIX EMCal measures energy deposits from particles generated in proton-proton (p+p) collisions
- Focus on decay photons from neutral pion (π^0) or eta meson (η)



- Analyze performance of the π^0 and η meson reconstructed invariant mass and resolution

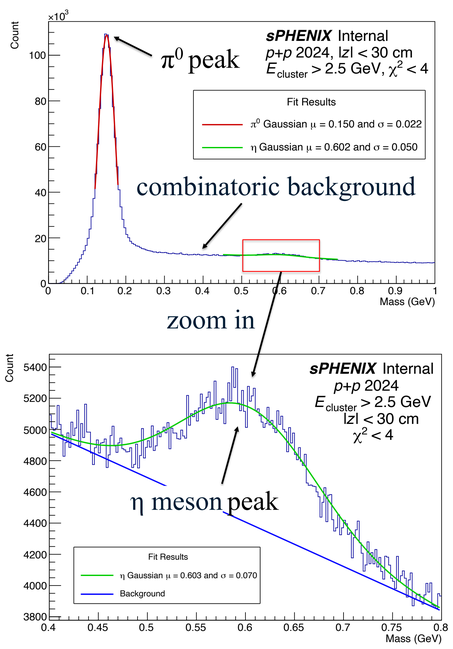


Methods

- Compare all possible cluster pairs to identify π^0 or η meson decay
- Calculate reconstructed invariant mass of the mother particle using cluster properties and kinematics
- Fit mass spectra with Gaussian

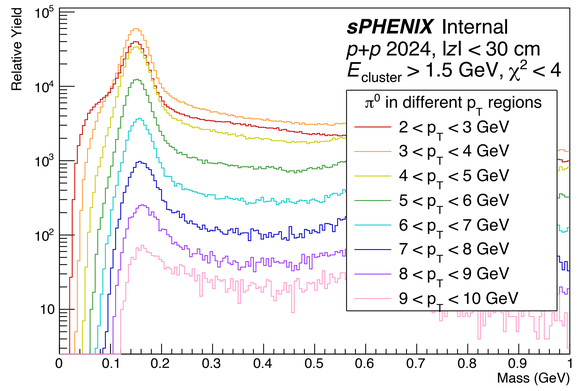
$$\underbrace{[0]\cdot\exp\left(-\frac{(x-[1])^2}{2\cdot[2]^2}\right)}_{\text{Signal}}+\underbrace{[3]+[4]\cdot x}_{\text{Background in }\eta\text{ fit}}$$

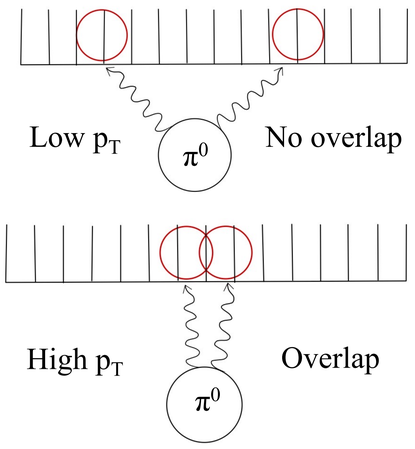
- Determine optimal cluster energy, shower shape quality, and vertex position thresholds
- Minimum Bias trigger only

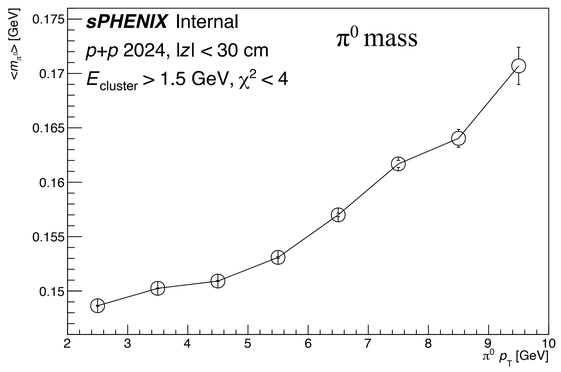


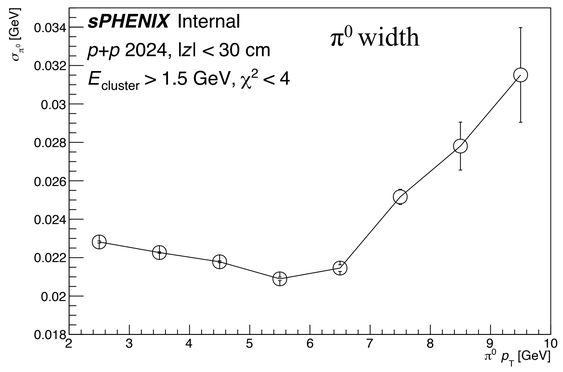
π^0 Invariant Mass and Relative Resolution as a Function of Transverse Momentum (p_T)

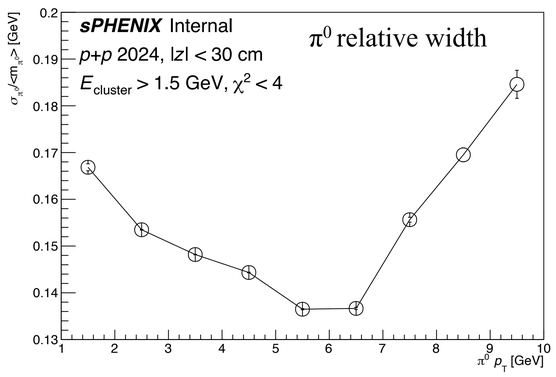
- Characterize π^0 mass peak position and width as function of energy to understand sPHENIX EMCal energy resolution
  - Test beam expectation: $\sigma/E = 2.8\% \oplus 15.5\%/\sqrt{E}$
- At high p_T, cluster pairs of photons with small angular separation begin to overlap due to segmented design of EMCal
- Effects of partial cluster merging visible starting around $p_T = 6$ GeV





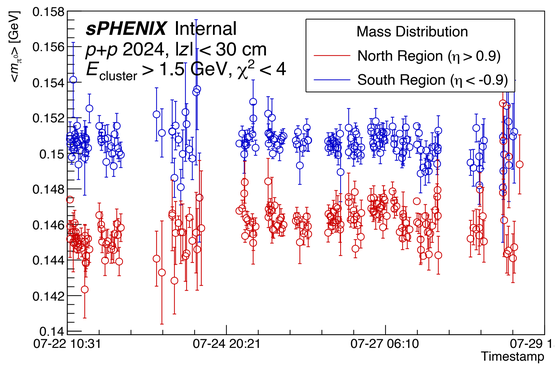


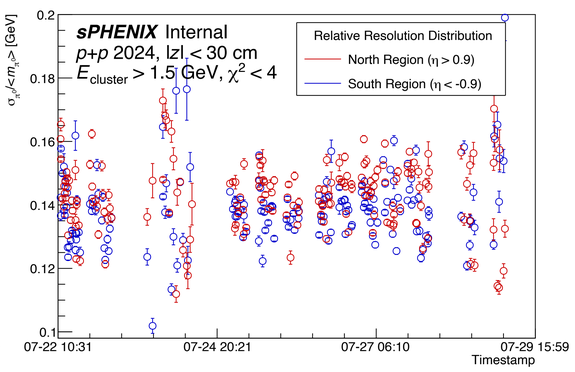




π^0 Invariant Mass and Relative Resolution as a Function of Time

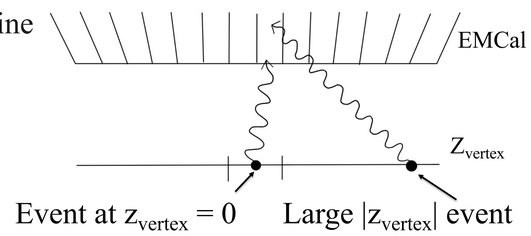
- Monitoring accumulating radiation damage in **North** and **South** regions of the EMCal
- EMCal is read out by silicon photomultipliers (SiPMs) that accrue radiation damage
- Larger beam backgrounds in **North** side of the detector, thus larger noise in SiPMs

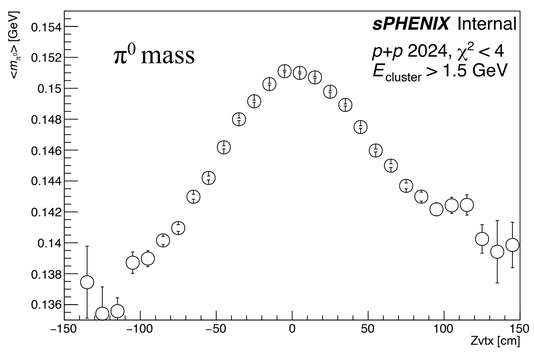


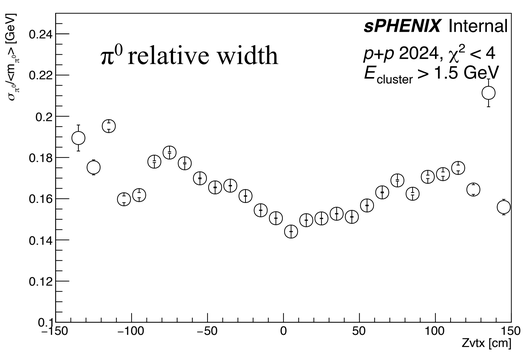


π^0 Invariant Mass and Relative Resolution as a Function of Collision Vertex (z_{vertex})

- Monitoring effects of changing RHIC machine conditions like beam-beam crossing angle
- Zero crossing angle data ($\sigma_{zvertex} = 55$ cm)
- Change in performance at larger $|z_{vertex}|$ due to EMCal becoming non-projective







Next Steps

- Analyze η meson mass and relative width as function of p_T, time, z_{vertex}
- Characterize full Run24 dataset for final EMCal calibration
- Determine a prescription for large $|z_{vertex}|$ events by optimizing clustering algorithm or implementing z_{vertex} dependent calibration

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References

[1] C.A. Aidala, et al., Design and Beam Test Results for the 2-D Projective sPHENIX Electromagnetic Calorimeter Prototype, *IEEE Trans.Nucl.Sci.* 68 (2021) 2, 173-181

[2] Belmont et al., Predictions for the sPHENIX physics program, *Nucl.Phys.A* 1043 (2024) 122821

[3] sPHENIX Technical Design Report, https://bit.ly/sphenix-tdr